Building AI Models for Embedding & Labeling scATAC-seq data from Human Peripheral Blood

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Introduction: What is scATAC-seq?

scATAC-seq (single cell Assay for Transposase-Accessible Chromatin) enables relatively unbiased identification of epigenetic activity at single cell-scale, but is hampered by sparsity and cell type identification. Recent advances allow for accurate open chromatin identification and modeling despite sparsity over 99%, while also enabling scRNA and ADT-based sequencing from the same cells. Specifically, the recent MOCHA package[1] enables cell-type specific peak calling and modeling in scATAC-seq, and MultiVI[2]/PeakVI[3] enables accurate embedding of scATAC-seq despite high sparsity. Further advances have been made by Marianno Gabitto's group in the development of CELLBLAST models for cell type labeling. However, these advances have not been combined to generate a high quality cell-type models for unpaired scATAC-seq.

For our work, we leverage the latest methodology advances, along with trimodal TEA-seq data[4,5] (Transcripts, Epitopes, & Accessibility), to generate cell-type-specific models for scATAC-seq data from human peripheral blood mononuclear cells.

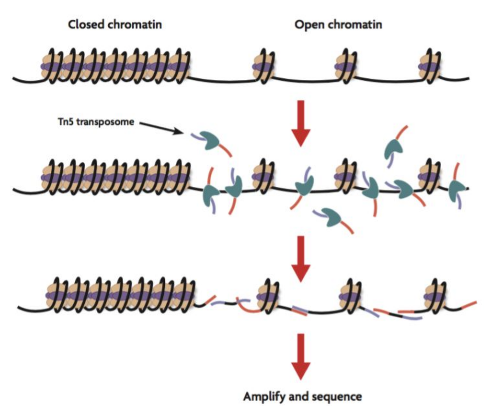

Figure 1: Image from ActiveMotif.com, reflecting how scATAC-seq protocol identifies open chromatin. https://www.activemotif.com/blog-atac-seq

Methods

1. Identify cell types in TEAseq data using standard scRNA-seq model
   - Embedding/UMAP done in Scanpy[6] using 5k variable genes
   - CellTypist models for cell labeling, trained on 13 million cells[7]
   - Cell label cleaning & confirmation via epitope data
2. Import labels in ArchR[8] and MOCHA for accurate peak calling
3. Identify cell type-specific peaks for cell type specific
   - Identify cell type-specific peaks in pseudobulk peak calls
   - Filter for least sparse regions
4. Conduct logistic regression to identify final feature set
   - Select top 1000 to 2000 features per cell types
5. Leverage Variational Autoencoders (VAE)
   - For embedding and label transfer

MOCHA Github

Results: Feature Identification & scATAC-seq Embedding

a) T cells | > 300,000 cells, 16 classes

b) B cells | > 10,000 cells, 5 classes

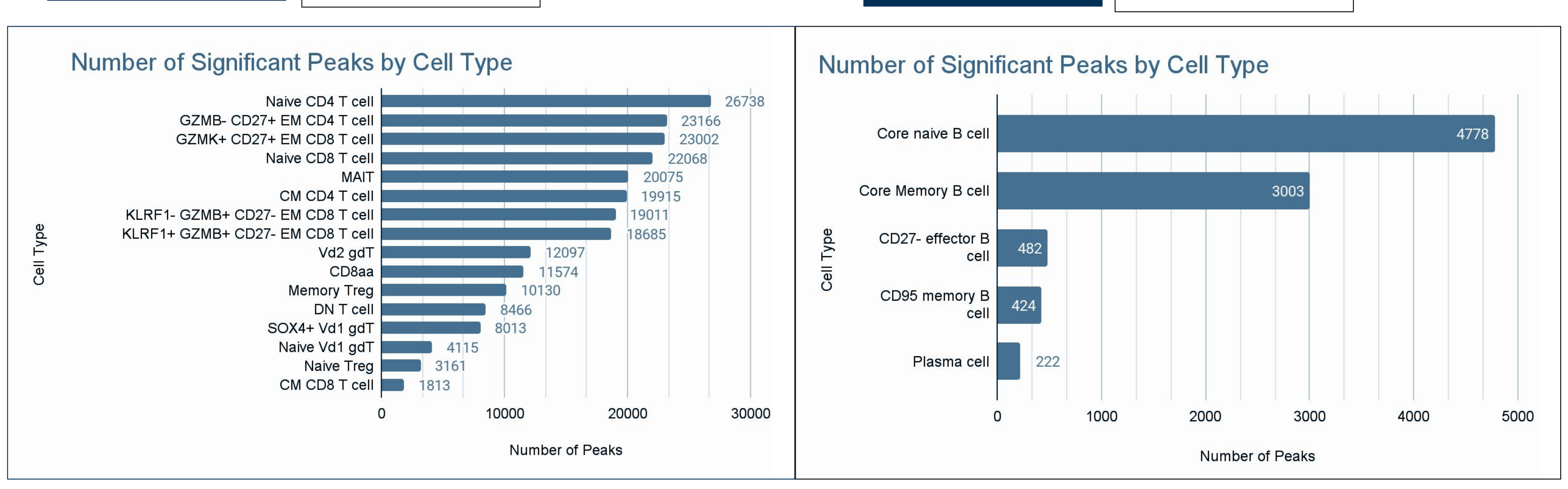

Figure 1: Identification of cell type-specific open chromatin downstream of MOCHA. a) The T cell data contained 323,357 cells over 16 cell types. b) B cell dataset contained 10,324 cells over five cell types.

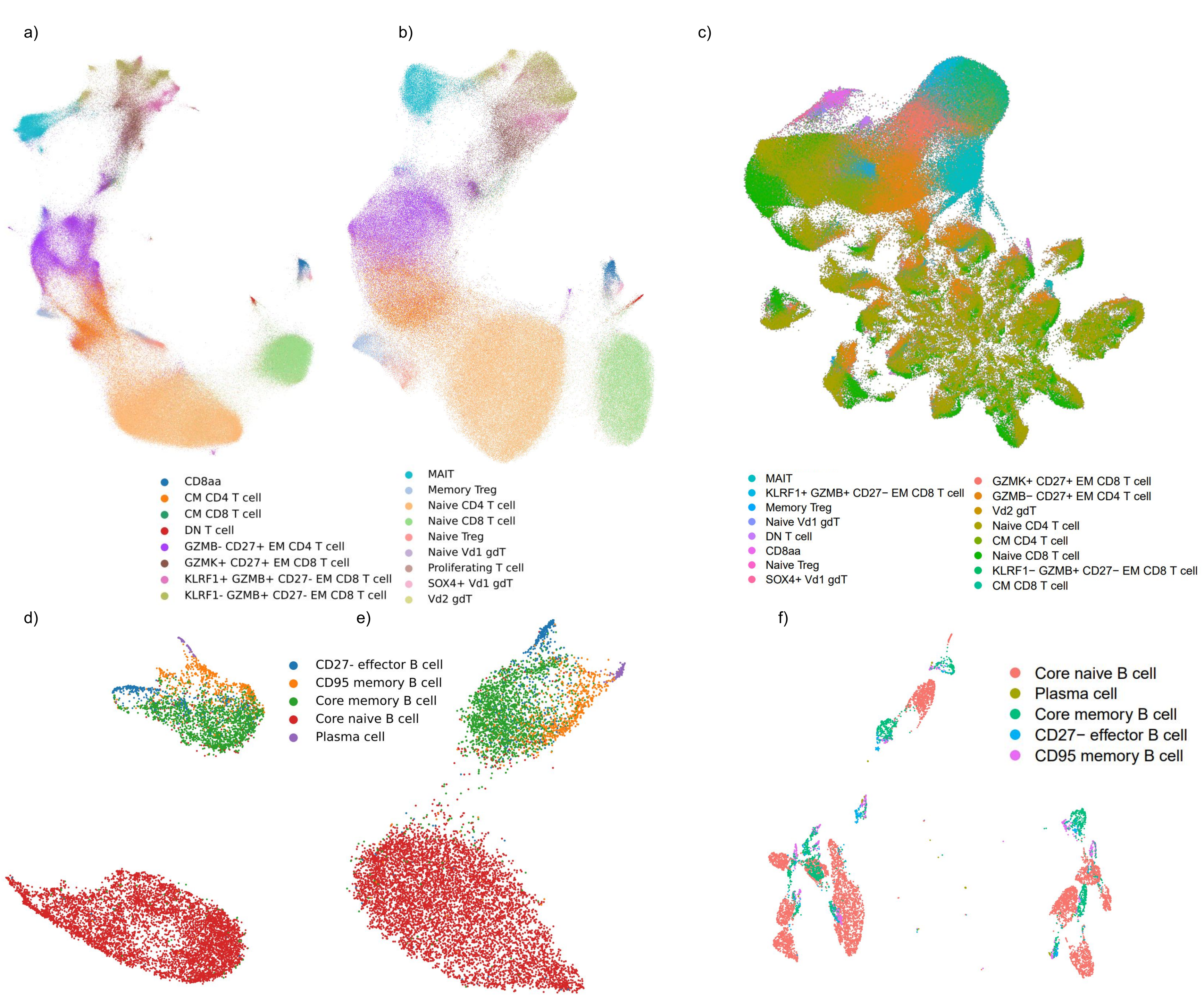

Figure 2: Select features recapitulate cell type embeddings within variational autoencoders (PeakVI), but not the more traditional, latent semantic indexing (LSI) approach. a-b) PeakVI-based UMAP of T cells over 40,133 & 14,582 features, respectively. Same legend for both. c) LSI-based UMAP of T cells over 14,582 features. d-e) PeakVI-based UMAP of B cells over 188,444 & 10,324 features, respectively. Same legend for both. f) LSI-based UMAPs of B cells using 10,324 cell type-specific features. Note that epigenetic states transitioning before transcriptomic ones within the PeakVI embeddings.

Results: Cell Type Classification for scATAC

CELL + CELLBLAST Training Latent Representation

CELL does initial training to generate an embedding space with clear separation between cell types. CELLBLAST takes a CELL MODEL, and conducts additional training with a classifier.

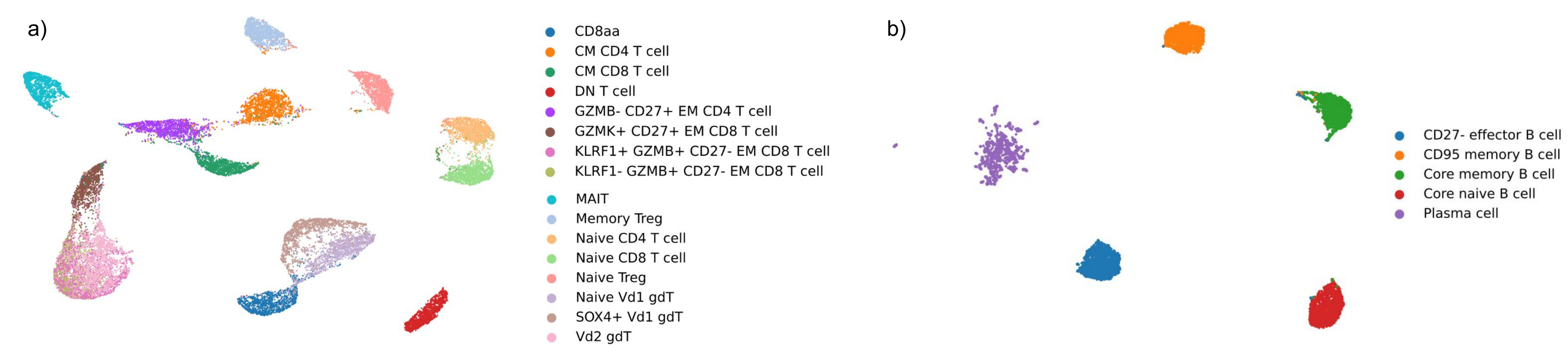

Figure 3: UMAP of CELL latent representation from the training datasets for a) T cells and b) B cells. CELL includes the ability to increase distances between cell types within the latent space. Only scATAC-seq data was used.

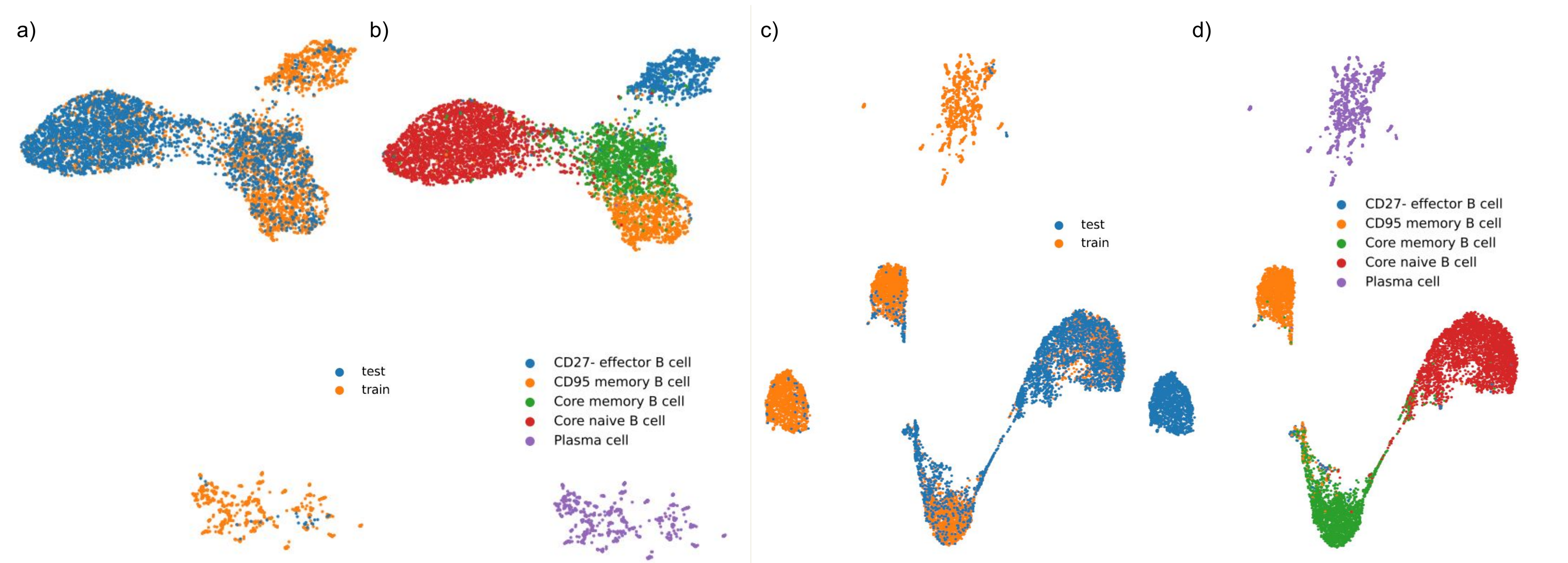

Figure 4: CELL & PeakVI embeddings of the training set and test set using the B cell dataset as an example. a-b) represent the PeakVI embedding generated on the training dataset, with the test set projected into it, color coded by test vs train labels, and by the ground-truth cell type labels, from left to right. c-d) represent the same for a CELL model, color coded by test vs train labels, and the ground truth labels, respectively. Only scATAC-seq data was used.

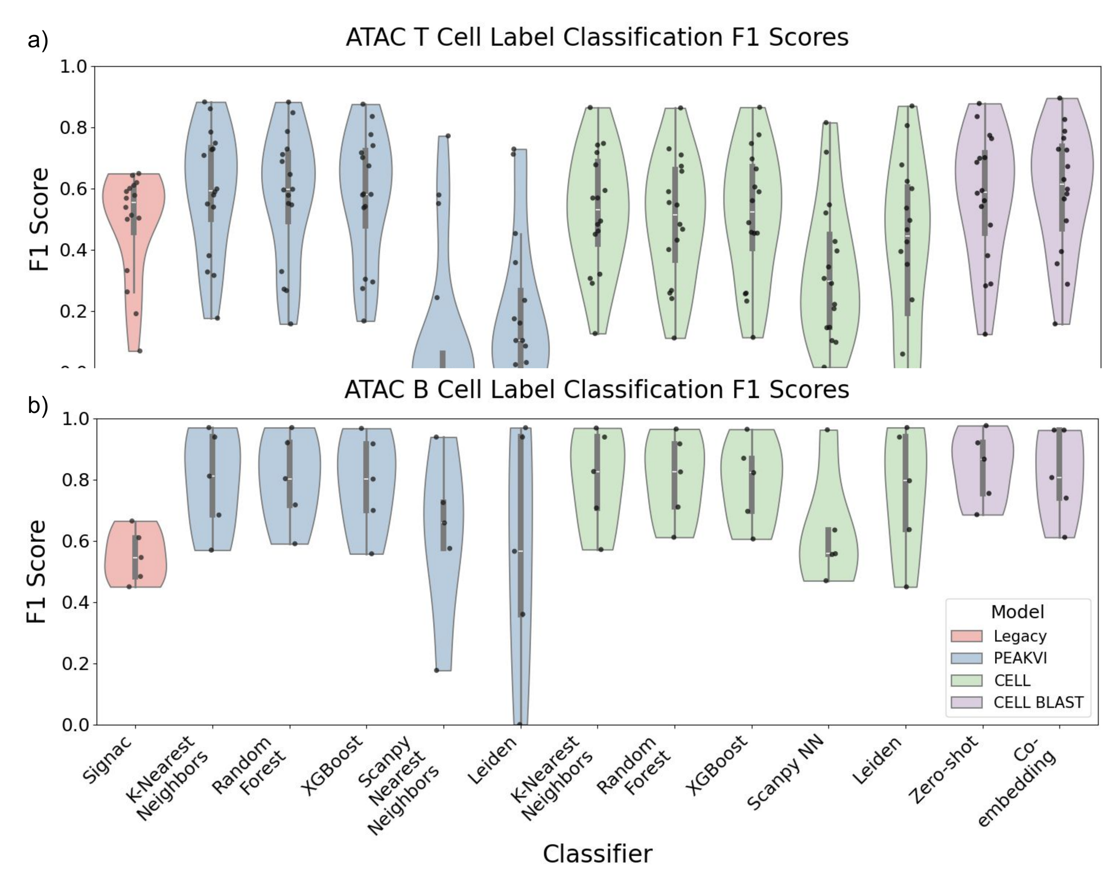

Figure 5: F1 labeling scores across a range of methods for the a) T cell and b) B cell datasets (scATAC only). The legacy approach is Signac's workflow (in red), leveraging an LSI embedding and Seurat's workflow via identify transfer anchors and mapping labels. Cell type classification based on PeakVI's latent space (blue) was conducted across a range of methods, including K-Nearest Neighbors, Random Forest, XGBoost, Scanpy's Nearest Neighbors, and Leiden clustering. Next, CELL's latent space (green) was used with the the same classification methods. Finally, CELLBLAST's classification VAE was used (purple). For CELLBLAST zero-shot prediction, the test data was held out entirely from training. For CELLBLAST co-embedding, the entire dataset was passed to CELLBLAST, but the test labels were withheld, and then later predicted based on the labels from the training set.

Results: Multimodal Analysis & Labeling

CELLBLAST on Multimodal Data

Running CELLBLAST using multiple modalities for co-embedding RNA & ATAC.

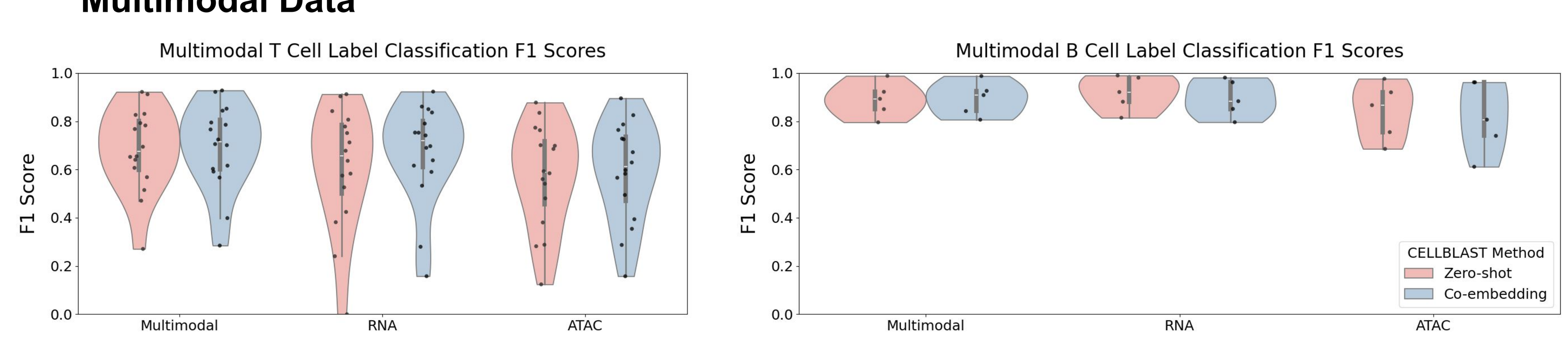

Figure 6: Violin plot of F1 scores from multimodal data from a) T cells and b) B cells using CELLBLAST, whether via zeroshot labeling with the test data withheld, or via co-embedding with the test labels withheld and then predicted.

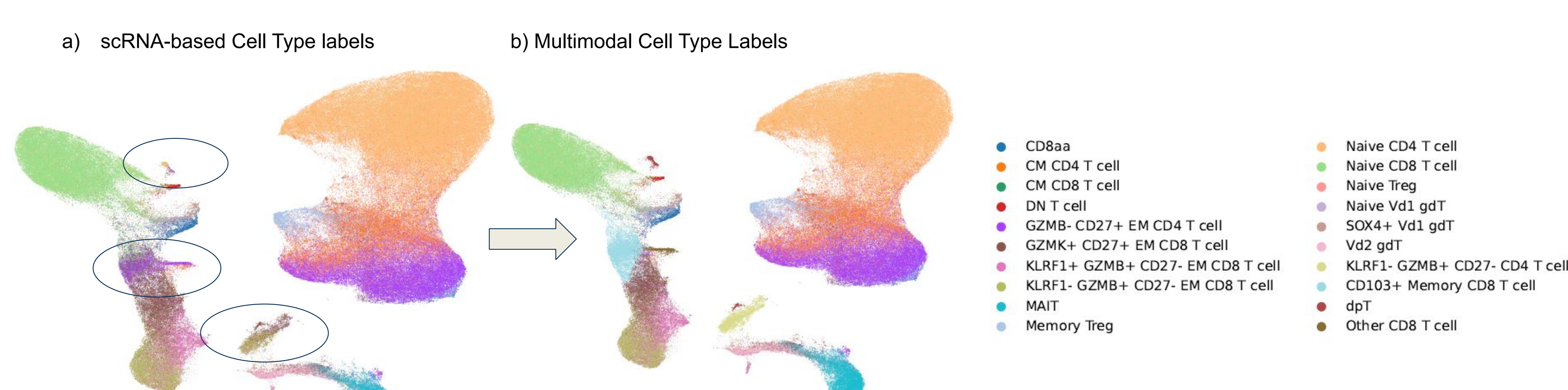

Figure 7: In-depth Multimodal analysis of T cell data reveals four additional distinct populations unclassified in scRNA or joint RNA-ADT analysis. a) Original MultiVI TEAseq UMAP with distinct cell types circled. b) Updated plot with double positive T cells (dpT), CD103+ Memory CD8 T cells, KLRF1- GZMB+ CD27- CD4 T cells, and fourth 'Other CD8 T cell' who's identify needs to be validated.

Next Steps

Next steps will focus on refining T cell labels to better reflect cellular heterogeneity, based on the trimodal MultiVi embedding. This tweak should increase cell type accuracy. After that, we plan to build a larger model across peripheral blood mononuclear cell types. On the model generation side, we will evaluate transfer learning methods for cell type classification, and further refine the CELLBLAST model to boost precision and recall. Finally, we plan to build trimodal labeling models for co-embedding/labeling scRNA and scATAC samples.

Acknowledgments

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Cell Typist Models & Immune Health Study

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